The Causal Firewall: Substrate-Independent Physical Constraints on the Origin and Distribution of Biological Complexity

Relational Actualism Paper IV

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Abstract

We derive substrate-independent physical constraints on the origin of biological complexity from Relational Actualism's causal graph dynamics, requiring no assumptions about chemistry, chirality, or aqueous solvents. The Causal Firewall is the Erdős–Rényi percolation transition at actualization density $\mu = 1$ — the same threshold governing QCD confinement, galactic rotation curves, quantum error correction fault-tolerance, cosmological seed viability, and the RA actualization selectivity maximum. Five independent physical scales are unified at a single dimensionless critical parameter. The minimum viable complexity for self-replication is $N_{\min} = 2$ causally connected actualization events — a non-Markovian structure with causal memory — identical to the minimum viable universe seed (Paper III). At biological temperatures ($T = 300\,\mathrm{K}$), the universal actualization timescale formula (Paper II) gives $\tau_d \approx 20\,\mathrm{fs}$, equal to the vibrational decoherence timescale; critically, this uses the RA on-shell selection criterion to identify the correct bath (optical phonons, not Debye acoustic modes) without additional assumptions. We establish two substrate-independent biosignature criteria: F1 (redundant environmental recording of actualization outcomes) and F2 (non-relativistic actualization rate), deriving them directly from the BDG dynamics rather than from chemical or biological assumptions. An explicit worked example traces the causal memory requirement from abstract graph topology through the BDG action to the minimum polymer length for a viable RNA self-replicator, bridging the formalism directly to prebiotic chemistry.

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1 Introduction: Dissolving the Contingency

The origin of biological complexity is conventionally treated as a problem of prebiotic chemistry: which molecules assembled, in which solvent, at which temperature? Different assumptions give different answers, and the entire inquiry is contingent on Earth-like conditions.

RA dissolves this contingency. The question is not "which chemistry?" but "what causal structure does self-replication require?" The answer is $N \geq 2$ causally connected actualization events — a non-Markovian structure with causal memory. This condition is derived from physics, not assumed from biology, and it applies regardless of the molecular substrate.

Substrate independence is the central claim, and the text will return to it repeatedly. Every section of this paper addresses the same question from a different angle: whether a given condition can be stated without reference to carbon, water, chirality, or any chemical specificity. Where we invoke specific chemistry (the RNA worked example, Section 4), we do so only to confirm the abstract prediction, not to derive it.

2 The Five-Scale $\mu = 1$ Unification

The single dimensionless parameter $\mu = 1$ (Planck actualization density) governs five independent physical scales:

1. **QCD confinement**: free quarks $\rightarrow$ color-confined hadrons at the Erdős–Rényi percolation of the color-charged graph.

2. **Galactic rotation curves**: transition from $d = 4$ to $d = 2$ effective graph geometry in sparse regions (Paper III, Section 3.3).

3. **Quantum error correction**: fault-tolerance threshold $p_{\text{th}} = \alpha_{\text{EM}} = 1/137$ (Paper V, Section 4.1).

4. **RA selectivity maximum**: D4U02 proves $\mu^* \approx 1.019$ **[DR]** (Paper I, Theorem 1).

5. **Causal Firewall**: minimum actualization density for a self-replicating causal structure to maintain identity against quantum decoherence.

These five appearances of $\mu = 1$ are the same geometric fact — the Erdős–Rényi percolation transition for the Poisson-CSG graph at Planck density — manifesting in different physical contexts. They are not independent occurrences connected by analogy; they are the same theorem applied at different scales.

3 The Correct Environmental Bath at $T = 300\,\text{K}$

3.1 On-Shell Bath Identification

The universal actualization timescale formula (Paper II, Eq. (3)) requires identifying which environmental modes are on-shell. There are two candidate baths in biological systems:

Optical phonon bath: characteristic frequency $\omega_{\text{opt}} \approx 3 \times 10^{13}\,\text{rad/s}$, correlation time $\tau_c \approx 20\,\text{fs}$, energy ratio $\hbar\omega_{\text{opt}}/(k_B T) \approx 0.76$ at $T = 300\,\text{K}$.

Debye acoustic bath: characteristic frequency $\omega_D \approx 10^{11}\,\text{rad/s}$, correlation time $\tau_c \approx 8.3\,\text{ps}$, energy ratio $\hbar\omega_D/(k_B T) \approx 3 \times 10^{-3}$ at $T = 300\,\text{K}$.

The RA on-shell criterion selects a bath mode as active if $\hbar\omega/(k_B T) \geq \Delta S^* \approx 0.6$. Optical phonons satisfy this at 300 K; Debye acoustic modes do not. This selection is a *derived* result from the threshold criterion, not an additional assumption. Standard decoherence theory (Lindblad) treats both baths equally; RA differentiates them by the threshold.

3.2 Derivation of τ_d

Applying the formula with the on-shell bath identified:

$$\tau_d = \frac{\Delta S^*}{\Gamma_{\text{opt}} \cdot (\hbar\omega_{\text{opt}}/k_B T)} = \frac{0.601}{3.93 \times 10^{13} \times 0.76} = 2.0 \times 10^{-14}\,\text{s} = 20\,\text{fs} \tag{1}$$

This equals τ_{vib}, the vibrational decoherence timescale. The result has no free parameters: ΔS^* is derived from the BDG model (Paper I), Γ_{opt} is determined by temperature, and the bath identification follows from the on-shell criterion.

3.3 Non-Markovian Correction

For prebiotic chemical environments with reorganization energy $\lambda_{\text{reorg}} \ll k_B T$ (slow-modulation limit), the Kubo line-shape theory gives a correction:

$$\kappa = \frac{\sqrt{2\Delta S^*}}{\mu_{\text{NM}}}, \qquad N_{\min}^{\text{NM}} = \left\lceil \frac{1}{\kappa} \right\rceil \tag{2}$$

For RNA-world conditions ($\lambda_{\text{reorg}} \approx 0.1$–$0.3\,\text{eV}$): $\kappa \approx 0.2$–0.5, giving $N_{\min}^{\text{NM}} \approx 2$–$5$. This is the range of minimum polymer lengths predicted by RA for the first self-replicating systems, consistent with known minimum RNA chain lengths for functional ribozymes ($\sim$ 10–30 nucleotides for minimal hairpin ribozymes [4]).

4 From Abstract Topology to Prebiotic Chemistry

4.1 The Causal Memory Requirement

The minimum viable complexity condition $N \geq 2$ connected actualization events is an *abstract* statement about causal graph structure. We now trace it to concrete molecular terms.

A self-replicating system requires:

1. An actualization event v_0 encoding chemical information (a specific molecular bond forming, closing a ring, setting a stereocentre).
2. A causally connected downstream event v_1 that uses the information from v_0 to reproduce the initial condition — i.e., a template-dependent reaction.
3. A directed edge $v_0 \prec v_1$ (causal memory): the outcome of v_0 constrains the outcome of v_1.

This is precisely $N = 2$ with a directed edge, giving $V = S = 1 - N_1 + 9N_2 - 16N_3 + 8N_4 = 1 - 1 + 9 = 9$. Without the directed edge ($N = 2$ disconnected), $V = 0$: the system cannot bootstrap.

4.2 The RNA Worked Example

Consider the minimal RNA self-replicating loop. An RNA strand acts as a template for its complement; the complement then templates the original sequence. In RA terms:

- v_0: nucleotide addition to the growing strand — an on-shell phosphodiester bond formation ($\Delta S > \Delta S^*$, irreversible in aqueous conditions).
- v_1: template-directed nucleotide addition to the complementary strand, causally constrained by the sequence established at v_0.
- Edge $v_0 \prec v_1$: the base-pairing specificity of the template is the causal connection.

The minimum strand length for this causal structure: a 2-nucleotide seed (the $N = 2$ connected graph) is kinematically sufficient for self-complementarity but too short to fold stably. The non-Markovian correction (Eq. 2) with $\kappa \approx 0.3$ for aqueous RNA conditions gives $N_{\min}^{\mathrm{NM}} \approx 3$–4. This corresponds to minimum RNA strand lengths of $\sim$ 6–8 nucleotides — consistent with the known minimum for stable RNA hairpin formation ($\sim$ 4–8 nucleotides).

This is not a chemical derivation. We derived $N_{\min} = 2$ from the BDG action and corrected it for the non-Markovian bath environment. The agreement with RNA-world minimum strand lengths is a *confirmation* of the abstract prediction, not the basis for it. A different chemical substrate (peptide nucleic acids, glycol nucleic acids, mineral-surface templates) with different κ would give a different minimum length — but the same abstract condition $N \geq 2$ with directed edge.

4.3 The Origin-of-Life Sandwich Bound

Any proposed origin-of-life mechanism must produce a system whose RA assembly depth A_{RA} lies between:

- Lower bound: $A_{\mathrm{RA}} \geq 2$ (non-Markovian causal memory with directed edge).
- Upper bound: $A_{\mathrm{RA}} \leq A_{\max}$ derived from the causal graph topology at the Planck scale.

The glycolysis pathway as a worked example: $A_{\mathrm{RA}} = 3$ (three non-trivial actualization depth levels, corresponding to the three main reaction stages — substrate phosphorylation, isomerization, and dephosphorylation). Glycolysis satisfies $2 \leq 3 \leq A_{\max}$.

RNA-world, metabolism-first, and crystal-template scenarios can each be evaluated against this bound using biochemical databases (KEGG/BiGG). The bound is substrate-independent: it constrains the causal depth structure, not the molecular identity.

5 Substrate-Independent Biosignature Criteria

5.1 F1: Quantum Darwinism Redundancy

Definition 5.1 (Condition F1)**.** *An environment satisfies F1 if it records the outcomes of actualization events redundantly enough to imprint them on macroscopic timescales — i.e., if the information content of actualization events is amplified into multiple environmental copies before the next actualization cycle.*

F1 makes no reference to carbon, water, molecular chirality, or any chemical substrate. It is a condition on the information-carrying capacity of the environment. Any medium satisfying the redundancy criterion qualifies: mineral surfaces, aerosol droplets, cosmic dust grains, ionic channels in exotic solvents.

5.2 F2: Non-Relativistic Actualization

Definition 5.2 (Condition F2). *An environment satisfies F2 if the local effective temperature T_{eff} satisfies $T_{\text{eff}} \ll T_U(a_{\text{loc}})$, where $T_U = \hbar a/(2\pi c k_B)$ is the Unruh temperature at local acceleration a_{loc}.*

In the extreme Rindler limit ($T_{\text{eff}} \gg T_U$), $\Delta S_{\text{per photon}} \to 0$ (by L05 **[LV]**, Paper II) and the actualization rate becomes uncontrolled. A system swept past a gravitational horizon cannot support stable biological complexity.

5.3 The Biosignature Prediction (P6)

$$\text{Life is found wherever F1 and F2 hold, regardless of chemistry.} \tag{3}$$

Environments to search beyond the conventional habitable zone:

- Subsurface rock-hosted systems: F1 via mineral-water interfaces; F2 via suppressed acoustic modes at $T \sim 40$–$80\,^{\circ}\text{C}$.
- High-pressure ices: F1 via ionic channels; F2 via low thermal noise.
- Interstellar grains: F1 marginal; F2 satisfied.

Environments where life is prohibited: extreme radiation environments (F1 destroyed), near stellar surfaces ($T_{\text{eff}} \approx T_U$ near neutron star surfaces — F2 violated), inside active galactic nuclei.

6 Discussion

The Causal Firewall provides a precise, parameter-free account of what biological complexity requires physically. F1 and F2 are physical theorems derived from the RA actualization criterion — they are not biochemical assumptions. The five-scale $\mu = 1$ unification demonstrates that the origin of life is not a special problem of chemistry but a particular manifestation of the same percolation physics governing QCD, gravitation, quantum computing, and the origin of universes.

The theory does not say which chemistry will produce life; it says *what any chemistry must produce* — a directed causal connection between at least two actualization events. The identification of which chemistry achieves this in a given environment is an application of the abstract constraint, not a derivation from it.

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